

Chapter 9: Ray Optics and Optical Instruments

Formula Sheet + Tips & Tricks

Formula Sheet

I–II. Spherical Mirrors

Mirror Formula

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

u = object distance, v = image distance, f = focal length ($f = R/2$), all measured from pole.

Magnification (Mirror)

$$m = -\frac{v}{u} = \frac{f}{f - u}$$

Negative m = real, inverted; positive m = virtual, erect.

III. Snell's Law and Refractive Index

Snell's Law

$$n_1 \sin i = n_2 \sin r$$

i = angle of incidence, r = angle of refraction, n_1, n_2 = refractive indices of the two media.

Refractive Index

$$n = \frac{c}{v} = \frac{\sin i}{\sin r}$$

c = speed of light in vacuum, v = speed in medium.

IV. Apparent Depth

Apparent Depth

$$n = \frac{\text{Real Depth}}{\text{Apparent Depth}}$$

Valid for near-normal viewing from a rarer medium.

Normal Shift

$$\Delta t = t \left(1 - \frac{1}{n} \right)$$

t = thickness of slab/liquid; Δt = apparent upward shift of the object.

V. Total Internal Reflection

Critical Angle

$$\sin C = \frac{1}{n}$$

n = refractive index of denser medium relative to the rarer medium; C = critical angle.

VI. Refraction at a Spherical Surface

Spherical Refraction

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

n_1 = incident-side index, n_2 = refracted-side index, R = radius of curvature of the surface.

VII–VIII. Lens Maker's Formula

Lens Maker's Formula

$$\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

n = refractive index of lens material relative to surrounding medium; R_1, R_2 = radii of the two surfaces.

Lens in a Liquid

$$\frac{1}{f_L} = \left(\frac{n_{\text{lens}}}{n_{\text{liquid}}} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

f_L = focal length of the lens when immersed in a liquid of index n_{liquid} .

IX–XI. Lens Formula, Magnification, Combination**Thin Lens Formula**

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

All distances measured from the optical centre.

Magnification (Lens)

$$m = \frac{v}{u} = \frac{f}{f + u}$$

Positive m = erect (virtual for convex, always virtual for concave); negative m = real, inverted.

Lenses in Contact

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$$

F = equivalent focal length of two thin lenses held in contact.

XII. Power of a Lens**Power**

$$P = \frac{1}{f \text{ (in metres)}}$$

Unit: diopetre (D). $P_{\text{total}} = P_1 + P_2$ for lenses in contact.

XIII. Combination of Lenses and Mirrors**Sequential Refraction/Reflection**

v_{final} found by applying lens/mirror formula stage by stage
Image of one element becomes the object for the next; effective mirror power $P_{\text{eff}} = P_{\text{lens}} + P_{\text{mirror}} + P_{\text{lens}}$ for lens-mirror silvered systems.

XIV–XVII. Prism**Prism Angle Relation**

$$A = r_1 + r_2$$

A = angle of prism, r_1, r_2 = angles of refraction at the two faces.

Deviation

$$\delta = i + e - A$$

i = angle of incidence, e = angle of emergence.

Refractive Index (Min. Deviation)

$$n = \frac{\sin \left(\frac{A + \delta_m}{2} \right)}{\sin \left(\frac{A}{2} \right)}$$

δ_m = angle of minimum deviation (occurs when $i = e$, $r_1 = r_2$).

Thin Prism Deviation

$$\delta = (n - 1)A$$

Valid for small prism angle A (thin prism approximation).

Angular Dispersion

$$\theta = (n_v - n_r)A$$

n_v, n_r = refractive indices for violet and red light.

XVIII–XIX. Microscopes**Simple Microscope**

$$M = 1 + \frac{D}{f} \text{ (image at } D\text{)}; \quad M = \frac{D}{f} \text{ (image at } \infty\text{)}$$

D = least distance of distinct vision (25 cm), f = focal length of lens.

Compound Microscope

$$M = m_o \times m_e = \frac{v_o}{u_o} \times \left(1 + \frac{D}{f_e} \right)$$

m_o = linear magnification of objective, m_e = angular magnification of eyepiece.

XX. Telescope**Astronomical Telescope**

$$M = -\frac{f_o}{f_e} \text{ (relaxed eye)}; \quad L = f_o + f_e$$

f_o, f_e = focal lengths of objective and eyepiece; L = tube length for image at infinity.

Telescope (Image at D)

$$M = -\frac{f_o}{f_e} \left(1 + \frac{f_e}{D} \right)$$

Used when the final image is formed at the least distance of distinct vision.

Tips, Tricks & Hacks

- Sign convention (Cartesian): distances measured against the incident light are negative, measured along it are positive; always fix the pole/optical centre as origin *before* substituting numbers — most mirror/lens errors come from skipping this.
- Mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ and lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ differ by a sign — write down which one you're using before you start, don't mix them mid-problem.
- For a concave mirror, remember the four zones by object distance vs f and $2f$: beyond C (real, inverted, diminished), at C (real, inverted, same size), between C and F (real, inverted, magnified), inside F (virtual, erect, magnified) — draw this number line once and you'll never forget it.
- Lens-maker's formula $\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$: if the lens is flipped or immersed in a denser medium, recompute n as the *relative* refractive index (lens w.r.t. surrounding medium), not the absolute one.
- Power of a lens $P = \frac{1}{f}$ (in metres) is additive for lenses in contact: $P = P_1 + P_2 + \dots$ — much faster than combining focal lengths directly, especially with more than 2 lenses.
- Critical angle and TIR: $\sin C = \frac{1}{n}$ — TIR only happens going from denser to rarer medium, and only when angle of incidence exceeds C ; this is the one-line justification for optical fibres, mirages, and sparkle in diamonds.
- Simple microscope magnification is $1 + \frac{D}{f}$ (image at near point) or $\frac{D}{f}$ (image at infinity) — the “+1” is the single most forgotten term in exams.
- Telescope magnifying power (normal adjustment) = $\frac{f_o}{f_e}$, and tube length = $f_o + f_e$ — for microscopes it's the objective's focal length that should be *small*, for telescopes it should be *large*; keep the two straight.